Odd finite projections and the even-accumulation lineability problem

Candidate substantial partial-result packet for arXiv:2303.03871

11 August 2026

Status: candidate substantial partial result, likely valid, subject to human review. For a three-dimensional space of bounded sequences, the problem reduces exactly to the parity of one-dimensional linear images of its finite joint cluster set $P \subset \mathbb{R}^3$. Such a space cannot lie in $L(2\mathbb{N}) \cup \{0\}$ whenever P is coplanar, centrally symmetric, has at most four points, or has a direction whose parallel-fiber count is odd. Thus any hypothetical three-dimensional counterconfiguration must be a noncentral, full-dimensional even set of at least six points satisfying an explicit even-fiber condition in every determined direction. The general lineability question remains open in this packet.

1 Source question and literature status

For a bounded real sequence $x = (x_n)$, let L_x denote its nonempty compact set of accumulation points, and for $A \subset \mathbb{N} \setminus \{1\}$ put

$$L(A) = \{x \in \ell_\infty : |L_x| \in A\}.$$

Menet and Papathanasiou ask in Question 1 on page 2 of arXiv:2303.03871 whether $L(2\mathbb{N})$ is lineable, meaning whether $L(2\mathbb{N}) \cup \{0\}$ contains an infinite-dimensional vector space [1]. The source page is reproduced in Appendix A.

A later preprint, arXiv:2511.08760, claimed that this set is not even three-lineable, but the current arXiv record is withdrawn and says that one argument needs clarification [2]. Its first-version proof tries to force one pair of finite joint cluster points to collide while avoiding all other pair-collision hyperplanes. This avoidance fails when another pair difference is parallel to the selected difference: the two collision hyperplanes then coincide. The results below isolate the exact finite projection statement needed for a repair and prove several broad cases without using the withdrawn step.

Idea of the proof

Given three bounded sequences, record every subsequential limit of the vector sequence (x_n, y_n, z_n) in a finite set $P \subset \mathbb{R}^3$. Taking a linear combination of the sequences is exactly the same as applying the corresponding linear functional to P: no accumulation points are lost or created. The functional can then be chosen geometrically. A normal to the affine span collapses a coplanar set to one point. Central symmetry produces an odd image by collapsing one antipodal pair to the center. For four affinely independent points, a generic functional orthogonal to one edge collapses exactly that edge. More generally, a functional orthogonal to a direction d and generic inside that plane collapses precisely the affine lines parallel to d.

2 The exact finite-cluster reduction

For a bounded vector sequence $u = (u_n)$ in $\mathbb{R}^r$, write

$$C(u) = \{p \in \mathbb{R}^r : u_{n_k} \to p \text{ for some strictly increasing sequence } (n_k)\}.$$

Lemma 1 (linear images of the joint cluster set)**.** *Let $x^1, \ldots, x^r \in \ell_\infty$, put $u_n = (x_n^1, \ldots, x_n^r)$, and set $P = C(u)$. If every L_{x^j} is finite, then P is a nonempty finite subset of $L_{x^1} \times \cdots \times L_{x^r}$ and, for every $a \in \mathbb{R}^r$,*

$$L_{\sum_{j=1}^r a_j x^j} = a(P) := \{a \cdot p : p \in P\}. \tag{1}$$

Proof. The vector sequence lies in a compact box. Hence P is nonempty, and each coordinate of a vector cluster point is a coordinate cluster point, proving the stated finite containment. If $p \in P$, a subsequence converging to p shows that $a \cdot p$ belongs to the left side of (1). Conversely, if a subsequence of $\sum_j a_j x^j$ converges, compactness gives a further subsequence on which u_n converges to some $p \in P$. The scalar limit is then $a \cdot p$. □

For a nonzero direction $d \in \mathbb{R}^3$, let $m_d(P)$ be the number of affine lines parallel to d that meet P. Only the direction, not the scale of d, matters.

Proposition 1 (odd parallel-fiber criterion)**.** *Let $P \subset \mathbb{R}^3$ be finite. If $m_d(P)$ is odd for some direction d determined by two points of P, then there is a nonzero $a \in \mathbb{R}^3$ for which $|a(P)|$ is odd.*

Proof. Choose a in the plane $d^\perp$. For a difference $e = p - q$ not parallel to d, the additional collision $a \cdot e = 0$ cuts out a proper line in $d^\perp$. Avoiding the finitely many such lines, we have $a \cdot p = a \cdot q$ exactly when $p - q$ is parallel to d. The fibers are therefore precisely the affine d-parallel lines meeting P, so $|a(P)| = m_d(P)$. □

When $|P|$ is even, $m_d(P)$ is odd exactly when an odd number of the d-parallel fibers have even cardinality. Indeed, the number of odd-sized fibers is even because their cardinalities sum to $|P|$.

3 Unconditional forbidden joint-cluster configurations

Theorem 1. *Let $x, y, z \in \ell_\infty$ be linearly independent and suppose every nonzero element of $M = \operatorname{span}\{x, y, z\}$ has a finite even number of accumulation points. Let*

$$P = C\big((x_n, y_n, z_n)_{n \geq 1}\big) \subset \mathbb{R}^3.$$

Then all of the following are necessary:

1. *P affinely spans $\mathbb{R}^3$;*
2. *$|P| \geq 6$;*
3. *P is not centrally symmetric;*
4. *$m_d(P)$ is even for every direction d determined by P.*

Equivalently, if any one of these four conditions fails, then M contains a nonzero sequence with an odd number of accumulation points.

Proof. Lemma 1 says that $|a(P)|$ is even for every nonzero coefficient vector a. A generic a avoids the finitely many hyperplanes $(p-q)^\perp$, so it is injective on P; consequently $|P|$ itself is even.

If P has affine dimension at most two, choose a nonzero a annihilating the linear span of $P-P$. Then $a(P)$ is a singleton, a contradiction.

Suppose P is centrally symmetric about c. Since $|P|$ is even, c is not in P. Pick $p \in P$, put $u = p - c \neq 0$, and choose nonzero $a \in u^\perp$. The finite set $a(P)$ is centrally symmetric about $a \cdot c$ and contains its center because $a \cdot p = a \cdot (2c - p) = a \cdot c$. Every noncentral value is paired with a distinct reflected value, so $|a(P)|$ is odd, again a contradiction.

It remains to justify the lower bound on $|P|$. If $|P| \leq 4$, the coplanar case is already excluded. Thus $|P| = 4$ and its points are affinely independent. Choose two points p_1, p_2 and put $d = p_2 - p_1$. No other pair difference is parallel to d: a shared endpoint would make three points collinear, while a disjoint parallel pair would give a nontrivial affine dependence among the four points. Proposition 1 applied generically in $d^\perp$ therefore gives exactly three image values.

Finally, Proposition 1 directly rules out odd $m_d(P)$. □

Corollary 1. *No three-dimensional subspace of $L(2\mathbb{N}) \cup \{0\}$ can have, for one (and hence any chosen) basis, a joint cluster set that is coplanar, centrally symmetric, has at most four points, or satisfies the odd parallel-fiber criterion of Proposition 1.*

4 The remaining finite-geometry lemma

The following statement would settle the source question negatively, and in fact would show that $L(2\mathbb{N})$ is not three-lineable:

> Every nonempty finite set $P \subset \mathbb{R}^3$ of even cardinality admits a nonzero linear functional a such that $|a(P)|$ is odd.

A sufficient strengthening is the corresponding planar assertion. A generic linear map $\pi : \mathbb{R}^3 \to \mathbb{R}^2$ is injective on a prescribed finite P, and a functional on $\pi(P)$ pulls back to one on P. In the plane, that stronger assertion says that some direction has an odd number of parallel lines meeting the finite set. The converse implication is unavailable: a planar set viewed inside $\mathbb{R}^3$ already has the constant one-point projection normal to its affine span.

This last parity assertion is not proved here. The collision-hyperplane argument alone cannot prove it: if $p_1 - p_2$ is parallel to $p_3 - p_4$, every functional collapsing the first pair also collapses the second. For example, the four points $0, e_1, e_2, e_1 + e_2$ already exhibit coincident pair-collision hyperplanes. This is the precise obstruction to the generic-avoidance step in the withdrawn preprint.

5 Verification, computation, limitations, and novelty

The proof was audited at three points: compactness gives the reverse inclusion in (1); the excluded sets inside $d^\perp$ are proper exactly for nonparallel differences; and a finite centrally symmetric image containing its center has odd cardinality even when additional collisions occur.

The accompanying script `code/check_small_projection_parity.py` exhaustively checked all 32,767 nonempty even subsets of the 4×4 integer grid. For each set it enumerated every determined direction and found an odd projection cardinality. It also checked all nonempty even subsets of the Boolean cube using the finite list of one- and two-direction kernel types. These computations found no counterexample, but they are not a proof of the remaining lemma and are not used in Theorem 1.

The run registry and all four cheap indexes were searched for the source id, the even-accumulation question, and the projection reduction. Bounded web searches through 11 August 2026 used the source title, exact question, the authors, arXiv:2511.08760, and finite-projection parity phrases. The only apparent exact later answer is the withdrawn preprint just discussed; no usable full literature resolution or these subcase theorems were found. This is not an exhaustive priority claim.

This packet does not answer whether $L(2\mathbb{N})$ is lineable and does not validate the withdrawn not-three-lineable claim. Its unconditional contribution is the exact finite reduction and the four forbidden classes in Theorem 1.

Human-review recommendation. Check Lemma 1 and the central-symmetry case first. Then verify that the generic choice in Proposition 1 excludes only nonparallel differences. The principal judgment is whether the four structural exclusions constitute a substantial enough partial result for promotion; the global parity lemma must remain explicitly open.

A Source-page evidence

References

[1] Q. Menet and D. Papathanasiou, *Structure of sets of bounded sequences with a prescribed number of accumulation points*, arXiv:2303.03871, 2023.

[2] A. Fovelle and J. B. Seoane–Sepúlveda, *Bounded sequences having an even number of accumulation points*, arXiv:2511.08760, withdrawn 16 November 2025; https://arxiv.org/abs/2511.08760.

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spaceable. When A is a subset of $\mathbb{N}\backslash\{1\}$, we can easily show that if A is cofinite then $L(A)$ is lineable. On the other hand, if A is a finite subset of $\mathbb{N}\backslash\{1\}$ then $L(A)$ is not lineable [7, Theorem 4.4]. If A is an infinite subset of $\mathbb{N}\backslash\{1\}$ which is not cofinite, Leonetti, Russo and Somaglia proved that both behaviors can appear. For instance, $L(2\mathbb{N}+1)$ is lineable while $L(\{3^n : n \geq 1\})$ is not lineable. In fact, they have shown that if $a_0 \geq 2$ and $a_{n+1} > 2a_n$ then $L(\{a_n : n \geq 0\})$ is not lineable. These different results have led to different open questions (see [7]).

Question 1. *Is $L(2\mathbb{N})$ lineable? Is $L(\{n^2 : n \geq 2\})$ lineable?*

Question 2. *Is $L(2\mathbb{N}+1)$ densely lineable in ℓ^∞?*

Question 3. *Does $L(\omega)$ or $L(\mathbb{N}\cup\{\omega\})$ contain a non-separable closed subspace? Does it contain an isometric copy of ℓ^∞?*

Excepted the case of $2\mathbb{N}$, we answer all these questions in the following sections. In Section 2, we focus on the lineability and the dense lineability of $L(A)$ for some subset $A \subset \mathbb{N}\backslash\{1\}$. More precisely, we prove that if $A \subset \mathbb{N}\backslash\{1\}$ and $L(A)$ is lineable then there exists $k \geq 1$ such that

$$|A \cap (A-k)| = \infty.$$

This allows in particular to deduce that $L(\{n^2 : n \geq 2\}$ is not lineable. We then prove that if $A \subset \mathbb{N}\backslash\{1\}$ and $L(A)$ is densely lineable in ℓ^∞ then

$$|A \cap (A-1)| = \infty.$$

This answers Question 2 negatively. Finally, we investigate the closed infinite-dimensional subspaces of $L(\omega)$ in Section 3 and we show that although $L(\mathbb{N}\cup\{\omega\})$ does not contain an isometric copy of ℓ^∞, the set $L(\omega)$ (and thus also $L(\mathbb{N}\cup\{\omega\})$) contains a non-separable closed subspace, answering Question 3.

2. Lineability and dense lineability

Leonetti, Russo and Somaglia [7] have already proved that $L(\mathbb{N}\backslash\{1\})$ and $L(\omega)$ are both densely lineable. They have also remarked that it follows from the results of Papathanasiou [11] that $L(\mathfrak{c})$ is densely lineable. For these reasons, we will restrict our study of lineability and dense lineablity to subsets A of $\mathbb{N}\backslash\{1\}$.

Given two sequences $x, y \in \ell^\infty$ with a finite number of accumulation points bigger than 2, we would like to better understand the possibilities of cardinality for the set $L_{\lambda x+\mu y}$ for any non-zero couple of real numbers (λ,μ). We will use in this section the notations introduced in [7]. In particular, as mentioned in [7], we can always find a partition $S_1,\dots,S_n$ of $\mathbb{N}$ in infinite sets and distinct real numbers $\xi_1,\dots,\xi_n$ such that $x \sim_{c_0} \xi_1 1_{S_1} + \cdots + \xi_n 1_{S_n}$, *i.e.* $x - \xi_1 1_{S_1} + \cdots + \xi_n 1_{S_n} \in c_0$ where $1_S = (z_n)_n$ with $z_n = 1$ if $n \in S$ and $z_n = 0$ otherwise. By considering a partition $T_1,\dots,T_k$ of $\mathbb{N}$ in infinite sets and distinct real numbers $\eta_1,\dots,\eta_k$ such that $y \sim_{c_0} \eta_1 1_{T_1} + \cdots + \eta_k 1_{T_k}$, we can easily state that

$$L_{\lambda x+\mu y} = \{\lambda\xi_i + \mu\eta_j : S_i \cap T_j \text{ is infinite},\ 1 \leq i \leq n,\ 1 \leq j \leq k\}.$$

In view of this equality, we introduce the following notations :

$$\mathcal{E}_{x,y} = \{(i,j) \in \{1,\dots,n\} \times \{1,\dots,k\} : S_i \cap T_j \text{ is infinite}\}$$

and

$$\mathcal{P}_{x,y} = \{(\xi_i,\eta_j) : (i,j) \in \mathcal{E}_{x,y}\}.$$

Figure 1: Page 2 of a local compilation of the official arXiv:2303.03871 source. Question 1 asks whether $L(2\mathbb{N})$ is lineable, and the following paragraph says this is the sole case not answered in the paper.